



Literature review over
microring modulators.



contents

- **Concepts.**
- **"A high-speed micro-ring modulator for next generation energy-efficient optical networks beyond 100 Gbaud." (Sakib, Meer, et al., 2021).**



contents



Concepts.





Concepts

- Voltage–Phase Relation:

In an electro-optic device, applying a voltage changes the refractive index of the material. That refractive index shift alters the optical phase of light traveling through the device.

Concepts

- Voltage-Phase Relation:

The change in index of refraction is phenomenologically described by:

$$\begin{aligned}\Delta n \text{ (at 1550 nm)} &= -8.8 \times 10^{-22} \Delta N - 8.5 \times 10^{-18} \Delta P^{0.8} \\ \Delta n \text{ (at 1310 nm)} &= -6.2 \times 10^{-22} \Delta N - 6 \times 10^{-18} \Delta P^{0.8}.\end{aligned}\tag{6.1}$$

The change in absorption is described by:

$$\begin{aligned}\Delta \alpha \text{ (at 1550 nm)} &= 8.5 \times 10^{-18} \Delta N + 6 \times 10^{-18} \Delta P \text{ [cm}^{-1}\text{]} \\ \Delta \alpha \text{ (at 1310 nm)} &= 6 \times 10^{-18} \Delta N + 4 \times 10^{-18} \Delta P \text{ [cm}^{-1}\text{]},\end{aligned}\tag{6.2}$$

Concepts

- Voltage–Phase Relation:

Then the effective index, **n_{eff}** as functions of applied voltage is given by:

$$n_{\text{eff}}(V) = n_{\text{eff},i} + \frac{\int E^*(y) \cdot \Delta n(y, V) E(y) dy}{\int E^*(y) \cdot E(y) dy} \cdot \frac{dn_{\text{eff}}}{dn_{\text{co}}}$$

where **n_{eff,i}** is the effective index of the waveguide without any doping and **dn_{eff}/dn_{co}** (change of mode effective index versus change in the waveguide core effective index) is typically very close to 1 (including silicon strip and rib waveguides). **E(y)** is the 1D field profile.

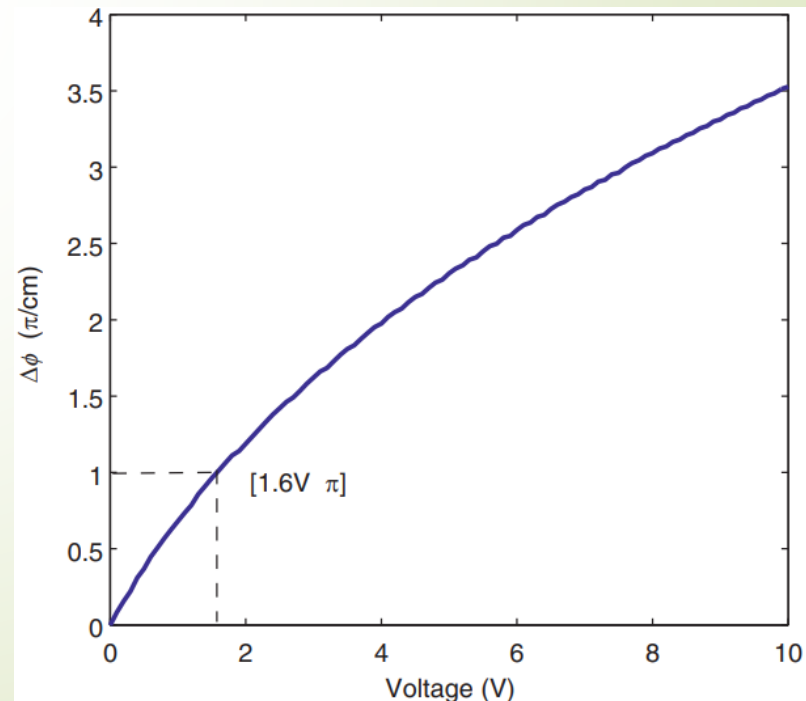
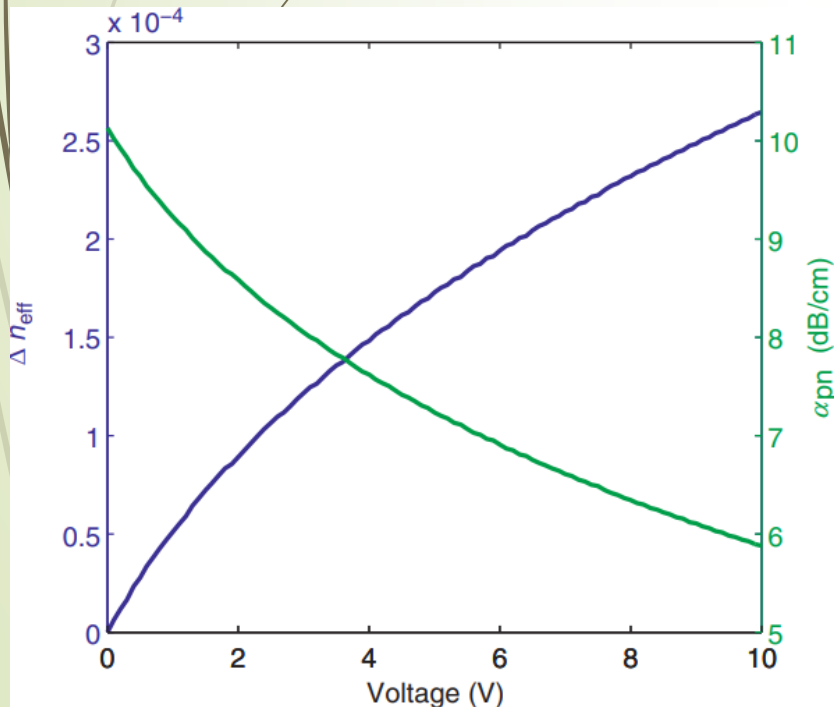
Concepts

- Voltage-Phase Relation:

Then the voltage-dependent changes in effective index and phase are given by:

$$\Delta n_{\text{eff}}(V) = n_{\text{eff}}(V) - n_{\text{eff}}(0)$$

$$\Delta \phi(V) [\pi \cdot \text{cm}^{-1}] = \frac{0.02 \Delta n_{\text{eff}}(V)}{\lambda}$$

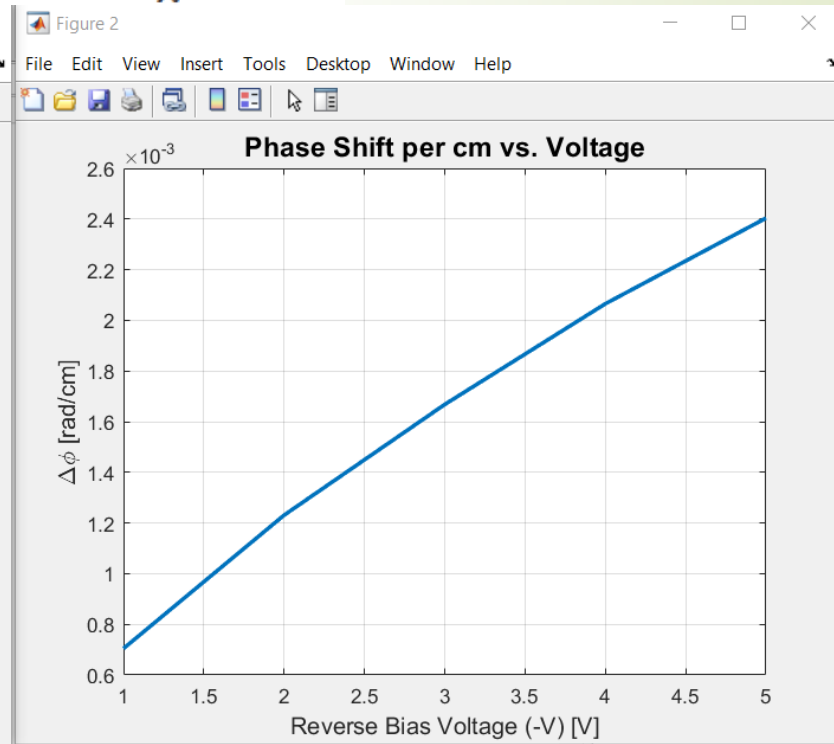
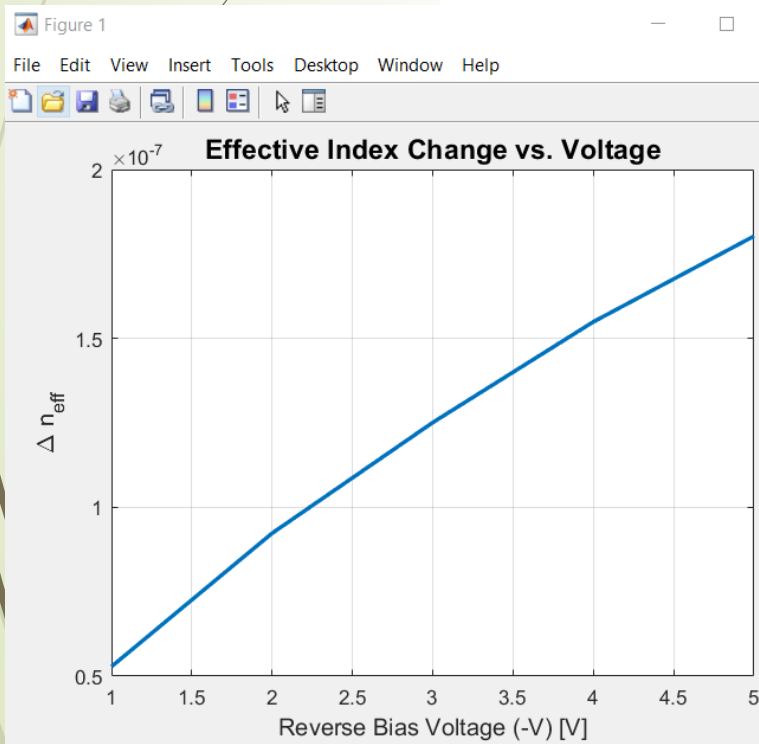


Concepts

- Voltage-Phase Relation:

Then the voltage-dependent changes in effective index and phase are given by:

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Concepts

- Voltage–Phase Relation:

it is important to note that:

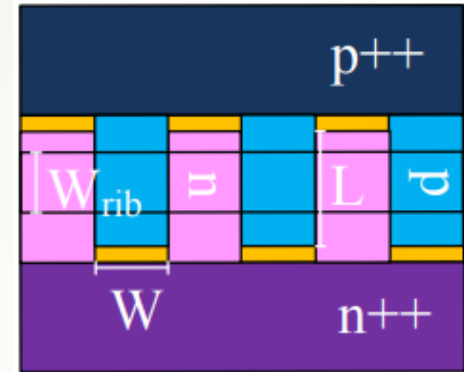
phase shift is non-linear with applied voltage.

phase shift in silicon saturates at higher voltages.

$$W_d = \sqrt{\frac{2\epsilon_0\epsilon_s(N_A + N_D)(V_{bi} - V)}{qN_A N_D}}$$

Concepts

- p-n junction configuration:
Interdigitated p-n junction:

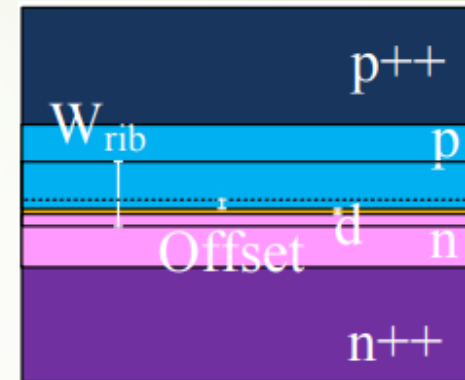


Offers better modulation efficiency, a $V_{\pi L \pi}$ lower by 40% , than the lateral p-n junction. This is because the interdigitated p-n junction completely overlaps with the optical mode.

This, however comes at the cost of higher capacitance which decrease the junction bandwidth. Hence, in terms of speed, lateral doping profiles was determined to be much better than the interdigitated junctions.

Concepts

- p-n junction configuration:
Lateral p-n junction:



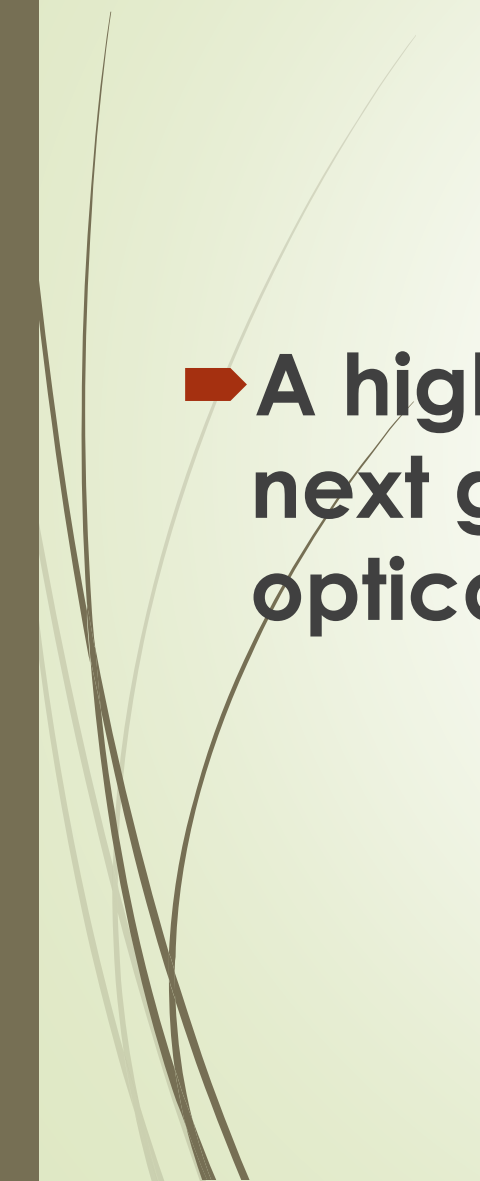
An optimal spacing (d) can reduce the capacitance, maintaining better modulation efficiency.

Concepts

- p-n junction configuration:
Vertical p-n junction:

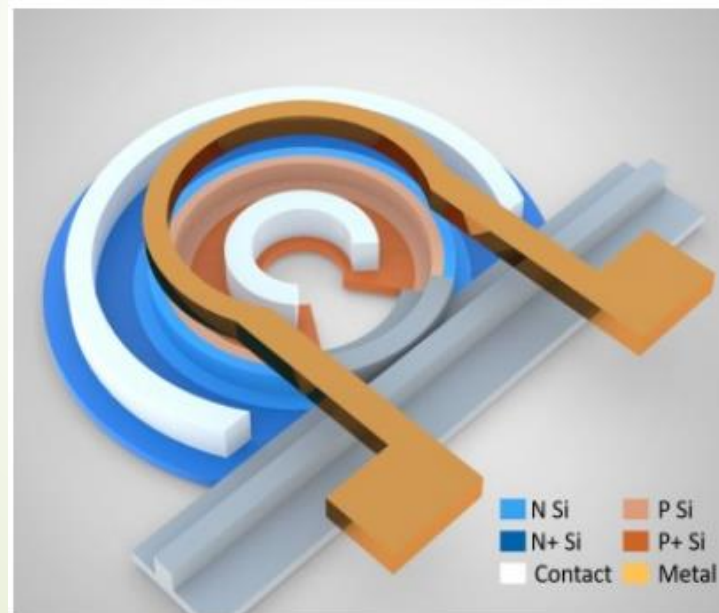


A vertical p-n junction has a greater overlap with the modulation region and the optical mode, thus it has a higher efficiency than a horizontal p-n junction. However, the vertical p-n junctions have a much smaller bandwidth due to a high RC constant.

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- **A high-speed micro-ring modulator for next generation energy-efficient optical networks beyond 100 Gbaud.**

➤ A high-speed micro-ring modulator for next generation energy-efficient optical networks beyond 100 Gbaud

- demonstrates a compact **6- μm** radius silicon microring modulator that achieves **128 Gbaud NRZ (0.8 V_{pp}, 3.8 dB ER)** with **≈ 5.3 fJ/bit** and also **192 Gb/s (96 Gbaud) PAM-4**.



A schematic of the silicon photonic MRM.

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- A high-speed micro-ring modulator for next generation energy-efficient optical networks beyond 100 Gbaud

- **Fabrication & geometry:**

Platform: SOI, rib waveguide (**300 nm** height, **400 nm** width, **100 nm** slab).

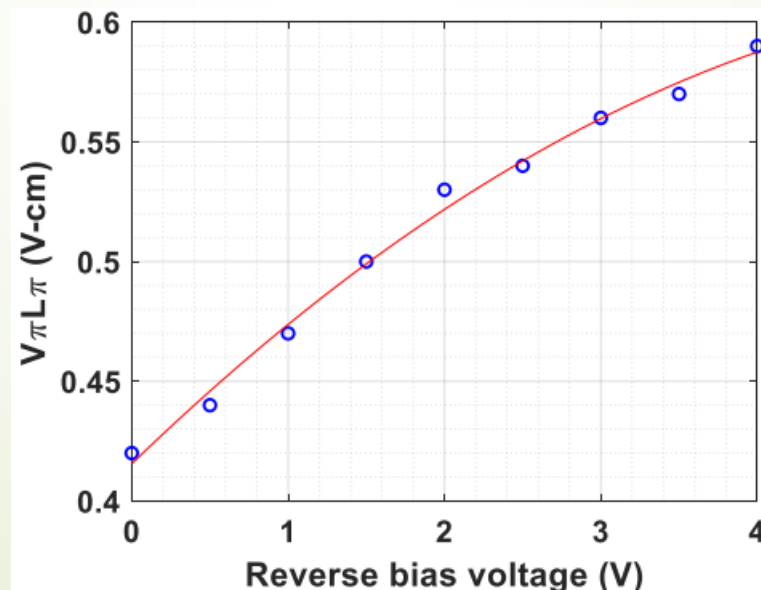
Ring radius: **6 μm** $\rightarrow$ FSR $\approx$ **11.3 nm** at 1310 nm.

Loaded Q: $\approx$ **4200**.

- A high-speed micro-ring modulator for next generation energy-efficient optical networks beyond 100 Gbaud

- Design & efficiency:

$V_{\pi} \cdot L$ (phase efficiency): **0.53** V · cm at 3 V reverse bias;
0.42 V · cm at 0 V

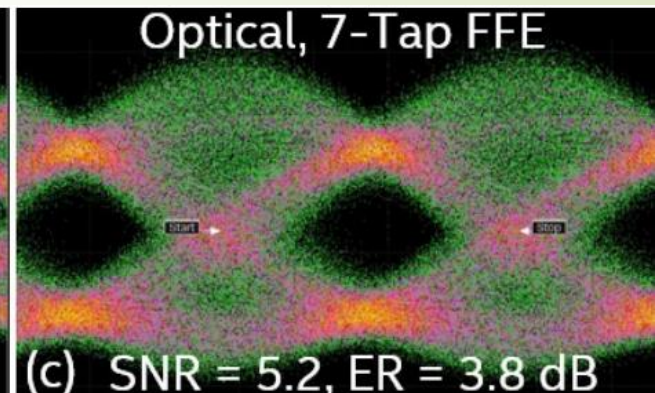
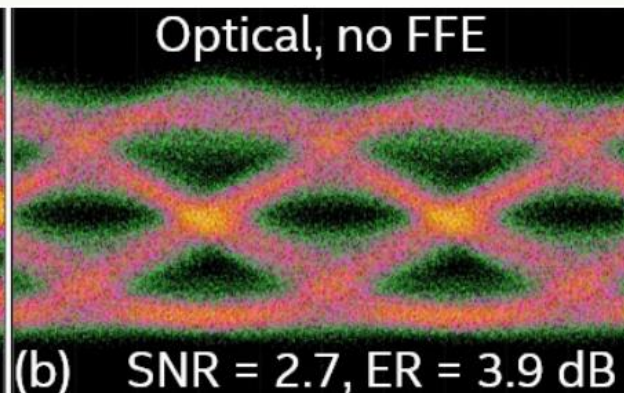
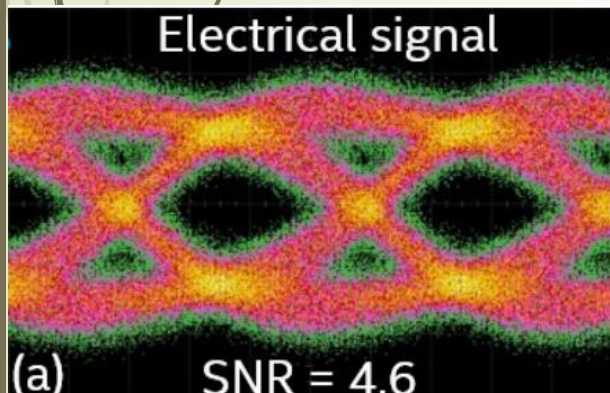


➤ A high-speed micro-ring modulator for next generation energy-efficient optical networks beyond 100 Gbaud

- Design & efficiency:

They obtained an open eye with SNR of **2.7** and ER of **3.9** dB, as shown in Fig.(b), without any equalization.

By applying a 7-tap feed forward equalization (FFE), the SNR can be improved to **5.2**, Fig. 4(c). Based on the eye SNR (Q-factor).





References

- Patel, David. Design, analysis, and performance of a silicon photonic traveling wave Mach-Zehnder modulator. McGill University (Canada), 2014.
- L. Chrostowski, M. Hochberg. Silicon Photonics Design From Devices to Systems, ch.6 modulators, Cambridge University Press, 2015.
- Sakib, Meer, et al. "A high-speed micro-ring modulator for next generation energy-efficient optical networks beyond 100 Gbaud." CLEO: Science and Innovations. Optica Publishing Group, 2021.